CAUSALITY ASSESSMENT

Pharmacovigilance Internship – Task 3 | Educational Case Analysis

Educational-use notice
All cases below are fictional practice examples, not real patient reports. Classifications are illustrative and based only on the facts stated. Do not use this document for clinical decisions or submit it as an actual regulatory safety report.

1. Objective

To evaluate the likelihood that a medicine caused a suspected adverse drug reaction (ADR) by reviewing the time relationship, known reaction profile, dechallenge (what happens after stopping the medicine), rechallenge information if it occurred, alternative causes and available clinical evidence.

2. WHO–UMC Causality Assessment

The Pharmacovigilance Programme of India (PvPI) follows the WHO–UMC causality assessment system. **Certain:** plausible timing, no reasonable alternative explanation, plausible response to withdrawal, and a definitive event; satisfactory rechallenge if necessary. **Probable/Likely:** reasonable timing, other causes unlikely, and clinically reasonable improvement after withdrawal; rechallenge is not required. **Possible:** reasonable timing, but disease or other medicines could also explain the event; withdrawal information may be missing or unclear. WHO–UMC also includes Unlikely, Conditional/Unclassified and Unassessable/Unclassifiable categories.

3. Case Study 1 – Immediate Hypersensitivity Reaction

Fictional patient: 30-year-old adult treated for a bacterial infection.
Suspected medicine: Penicillin antibiotic (specific product and dose not supplied in this training scenario).
Timeline: Within 20 minutes of administration, the patient developed widespread hives, lip swelling and breathing difficulty. Emergency care was provided. No other new medicine or clear alternative trigger was identified in the fictional case.
Assessment: The close timing and characteristic clinical presentation support a strong medicine relationship. However, a real case requires verified clinical assessment and complete records.
Training classification: Certain – illustrative only.

4. Case Study 2 – Dry Cough with Enalapril

Fictional patient: 56-year-old adult receiving treatment for hypertension.
Suspected medicine: Enalapril.
Timeline: Persistent dry cough began about two weeks after enalapril was started. The clinician considered other likely causes and changed the treatment. The cough gradually resolved. No rechallenge was performed.
Assessment: The reasonable timing and improvement after clinician-directed withdrawal/change support a drug relationship; other causes were considered unlikely.
Training classification: Probable/Likely. Rechallenge is not required and should not be attempted solely to prove causality.

5. Case Study 3 – Gastrointestinal Symptoms During Metformin Treatment

Fictional patient: 44-year-old adult recently started on metformin for type 2 diabetes.
Suspected medicine: Metformin.
Timeline: Nausea and loose stools began a few days after treatment started. The patient also had a recent dietary change and a family member with a short-lived gastrointestinal illness. Symptoms fluctuated; no clear dechallenge information was available.
Assessment: The timing is compatible with a possible medicine-related reaction, but alternative causes are also plausible and withdrawal response is unclear.
Training classification: Possible.

6. Summary Table

Case	Suspected medicine	Reaction	Classification
1	Penicillin antibiotic	Hives, lip swelling, breathing difficulty	Certain*
2	Enalapril	Persistent dry cough	Probable/Likely
3	Metformin	Nausea and loose stools	Possible

*Illustrative classification for this fictional scenario only.

7. Factors to Check During Causality Assessment

• Time relationship: Did the event begin after the medicine was started, with a plausible interval?

• Dechallenge: Did the event improve after the medicine was stopped or reduced?

• Rechallenge: Did it recur after re-exposure? Never intentionally re-expose a patient just to test causality.

• Alternative causes: Could the disease, another medicine, infection or another factor explain the event?

• Known reaction profile and objective evidence: Is the event consistent with known pharmacology, examination findings or laboratory results?

• Completeness: Are dose, route, dates, concomitant medicines, medical history and outcome documented?

8. Conclusion

Causality assessment is a structured, provisional judgement based on the available evidence. Certain requires strong evidence and no reasonable alternative explanation; Probable/Likely has reasonable timing and a supportive withdrawal response with other causes unlikely; Possible has reasonable timing but other explanations remain. The category may change when follow-up information becomes available.

9. References

1. Indian Pharmacopoeia Commission, PvPI Toolkit – WHO–UMC causality assessment: https://ipc.gov.in/PvPI/toolkit.html
2. WHO, The use of the WHO–UMC system for standardised case causality assessment: https://www.who.int/publications/m/item/WHO-causality-assessment
3. CDSCO, Guidance Document for Marketing Authorization Holders – Appendix on WHO–UMC causality assessment.

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